Learning with Less: Comparing Word and Character Features for News Classification in Four Nigerian Languages

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Abstract

How do word and character features compare across different amounts of labelled training data? We examine this question using MasakhaNEWS article bodies in Yorùbá, Igbo, Hausa and Nigerian Pidgin.

For each language, we compare logistic regression classifiers with word unigram and bigram TF-IDF against character 3-5-gram TF-IDF under fixed settings. Text normalisation and word tokenisation preserve diacritics. After overlap filtering, both representations use matched training samples at approximately 25%, 50% and 100% of the retained data and a fixed test set. Five class-stratified samples are used at the smaller sizes; each full-data model is fitted once. TF-IDF vocabularies and weights are learned only from the corresponding training sample.

Both approaches improve as training data increases. Character features achieve higher mean macro-F1 in all twelve language-size comparisons, with gains of 1.11-3.05 points on a 0-100 scale, although the advantage does not consistently widen at smaller training sizes. At full training size, character versus word scores are 87.90 versus 86.79 for Yorùbá, 72.63 versus 69.94 for Igbo, 86.67 versus 85.26 for Hausa, and 84.86 versus 82.75 for Nigerian Pidgin. Paired test-article bootstrapping with 10,000 resamples gives respective 95% percentile intervals for the character-minus-word differences of [-0.87, 3.12], [-1.89, 7.62], [-0.61, 3.44] and [-1.65, 6.74]. All include zero. Per-class gains are uneven, with some large improvements occurring in small categories.

These exploratory findings support further evaluation of character TF-IDF as a baseline, while leaving the full-data advantage uncertain. The comparisons share data, and the intervals capture test-sample uncertainty conditional on fitted models, not training variability. Separate hyperparameter tuning is needed to assess whether the observed differences persist. Evaluation beyond news remains future work; differing dataset sizes and categories preclude interpreting scores as a ranking of language difficulty.